The Cathedral:

A Machine-Checked Roadmap to the Riemann Hypothesis

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Abstract

We describe a Lean 4 formalization that reduces the 167-year-old Riemann Hypothesis to exactly two mathematical statements. A computer has verified that if both statements are true, the Riemann Hypothesis follows. This note explains the result, the two remaining obstacles, and why they matter—without requiring expertise in formal verification or analytic number theory.

1 What Did We Do?

The Riemann Hypothesis (RH) is one of mathematics' deepest unsolved problems. It concerns the distribution of prime numbers and has resisted proof since 1859.

We used a computer proof assistant called **Lean 4** to check every logical step of a proof attempt, starting from two precisely stated assumptions and ending at RH. The computer verified 3,444 separate modules with zero errors and zero gaps.

We did not prove RH. We identified exactly two mathematical claims that, if true, would imply RH—and we proved everything else with machine-checked rigor.

2 The Two Remaining Pieces

2.1 Piece 1: "The Sieve Bound"

Imagine a large matrix whose entries measure how similar different number-theoretic functions are to each other. RH is equivalent to saying that certain averages of these entries stay well-behaved.

Our first assumption says: the total "excess" in the off-diagonal entries of this matrix grows at most linearly with the matrix size. Numerical experiments with matrices up to size 100,000 confirm this with an 18× safety margin—the actual values are negative.

Why we can't prove it yet: It requires number-theoretic estimates (related to how greatest common divisors distribute) that are not yet available in the Lean mathematics library.

2.2 Piece 2: "The Spectral Wall"

If RH were *false*, there would be a "rogue zero" of the Riemann zeta function sitting in the wrong part of the complex plane. Our second assumption says: such a rogue zero would create an impenetrable wall—no matter what combination of basis functions you try, you can never approximate the target function closely enough.

This is known to be true by a classical theorem from the 1950s (Nyman–Beurling), but the proof uses deep tools (the Prime Number Theorem and Möbius function analysis) that are not yet formalized in Lean.

Why we can't prove it yet: The proof requires the Prime Number Theorem to be available in Lean 4, which is a major formalization project in itself.

3 What Did We Learn?

During this work, we made two surprising discoveries:

1. **The Sawtooth Floor:** A bound that seemed obviously true in informal mathematics turned out to be *false* for large values. The computer's insistence on rigor led us to find a subtle constant-covariance effect that invalidates the naive approach.

2. **The Hyperplane Trap:** A natural strategy for proving Piece 2 — using the Cauchy-Schwarz inequality — fails for a beautiful geometric reason: the "cheating" weights that would fool the inequality cause the approximation to explode in size. The computer warned us that the approach was doomed before we wasted weeks pursuing it.

Both discoveries show that formal verification is not just a way to check known proofs—it actively reveals errors that informal reasoning misses.

4 How to Verify

```
git clone https://github.com/jrgochan/prime
cd prime/proofs
lake build
```

The build completes in approximately two minutes and reports zero errors.